

The AI layer is the core part. It understands the intent and structures, and triggers the workflow to capture the required data. It applies suitable AI algorithms to analyse the data, the results of which are communicated back to the user.

IT services are complex and composed of many heterogeneous components. To integrate with these components, virtual assistants must implement different types of interfaces. The widely used interfaces are PowerShell, REST, OData, SQL, SNMP, etc.

Table 2 summarises the AI algorithms that are suitable for responding to typical inquiries that get triggered during a typical IT service root cause analysis and IT operations. The AI algorithms mentioned are indicative. There are a lot of algorithms available in each segment. We need to choose the right algorithm based on context. A detailed explanation of AI algorithms is beyond the scope of this article. Tons of reference literature is available publicly about AI algorithms.

Figure 2 gives an illustration of how to build a digital assistant for a virtual desktop service. This service consists of many layers, and each one has different types of interfaces. Digital assistants should implement interfaces to interact with all these layers.

Advanced auto-heal AI assistant

Digital assistants can be extended to detect, analyse and auto-heal issues arising during regular operations. In this case no human interaction is needed. Figure 3 shows logical layers of the auto-heal assistant. We need to use many AI algorithms to analyse the issues and trigger correct resolutions. We can make use of neural networks to learn and adapt to dynamic changes.

State/Condition: This part defines the rule. The rule is evaluated and if the condition is true, then the state

Table 2

Inquiry	AI algorithms in scope
Translate utterances to intents (Conversational engine)	NLP
When X event occurred in server N, how was the dependant storage service performance?	Log, event analysis
Data stream analysis	Statistical analysis
Identify causes of issues by calculating highest probability of the issue occurring	Bayesian algorithm
Prioritise root cause options	Bayesian algorithm
Find anomaly patterns in resource utilisation	Time series analysis (e.g., ARIMA)
Classify issues	Text analysis
Find out optimal resource groups	Segmentation (e.g., K-Means)
IT operations accelerators	Automation

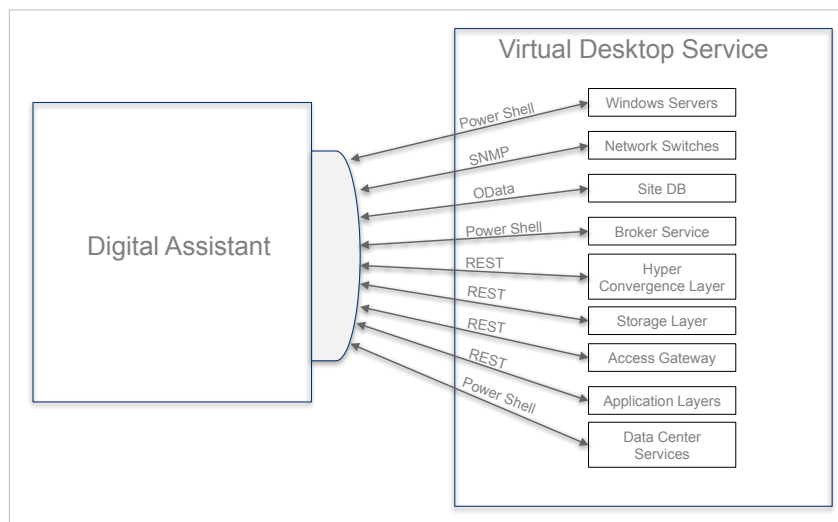


Figure 2: Digital assistant for virtual desktop service

is generated. State can be evaluated when a condition occurs, and is checked on a periodic basis. For example, when an application or system event occurs, then it will be evaluated. If the condition is evaluated as true, the state is processed. In another case, the performance metrics are checked on a periodic basis (for example, once in a minute), the condition is evaluated and state is generated based on the result.

Action: This part defines what action is to be taken in case the trigger occurs. Typically, these are remedial actions (such as restarting a service, increasing resource allocation, etc) to rectify an issue and heal it automatically without human intervention. In case self-healing is not available or it fails, then the administrator can be notified and the service tickets raised automatically.